

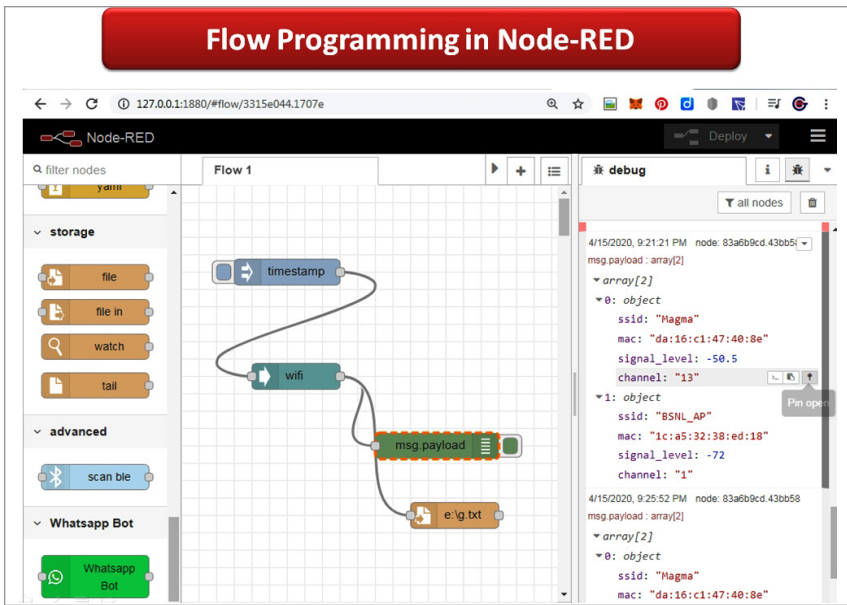


Fig. 6: Creating a flow diagram in Node-RED with Wi-Fi interfacing

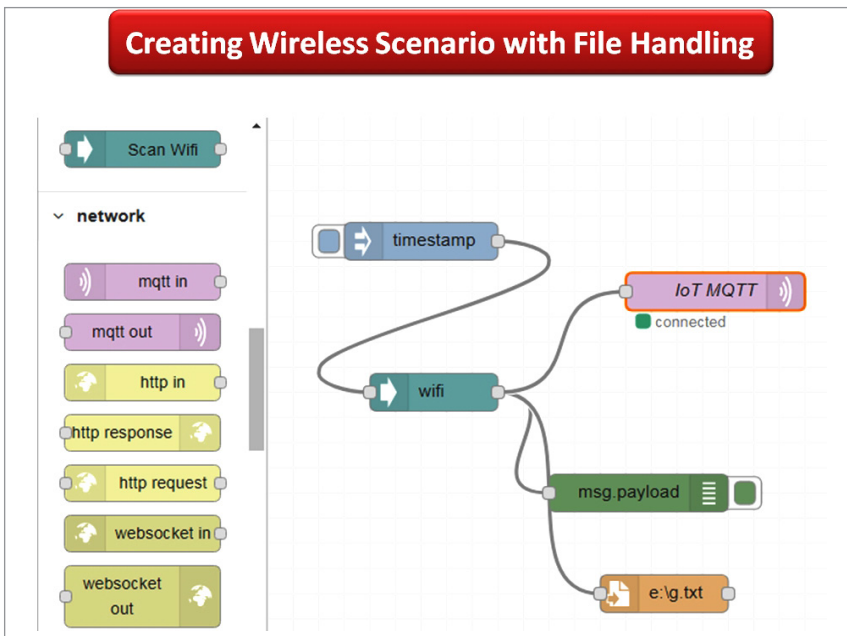


Fig. 7: Creating an IoT scenario for real-time interfacing with MQTT broker

for the Node platform (Fig. 2).

In the installation directory of Node.js, the `npm` command is used to install Node-RED, as follows:

```
E:\>cd nodejs
E:\nodejs>npm i node-red
```

Node-RED can be installed on multiple platforms, including open source hardware devices, IoT devices, the cloud, or on dedicated

servers. Its cloud deployment can be done on IBM Cloud, Microsoft Azure, SenseTecnica FRED, and Amazon Web Services. To work with IoT, fog and edge scenarios, it can be installed on Raspberry Pi, BeagleBone Black, or Arduino. These devices support the interfacing of IoT sensors for multiple applications.

To run Node-RED on a dedicated

system, the `node-red` command is executed to start the server. A local host based IP address with port number is generated, on which the programming for IoT, edge computing, etc, is done.

After starting the server, there are thousands of packages that can be interfaced for multiple applications. To create an IoT or edge scenario, packages related to Wi-Fi, Bluetooth, MQTT, CoAP, weather forecasting, etc, can be installed from the NPM repository.

After the required packages are installed, the flow diagram is created using the left palette of icons. Here, all the packages are visible and can be dragged across the dashboard.

The message queueing telemetry transport (MQTT) protocol is available in the Node-RED palette to help it work with IoT. It is used as an IoT broker for real-time interfacing and communication with sensors. An IoT scenario is created by the file handling module, capturing data from sensors and using the MQTT protocol to interface it with Wi-Fi. After running this scenario on a single click, the results are visible on the right side panel of Node-RED. Similar other structures can be created and deployed for dynamic communication with IoT devices.

Researchers and engineers can create IoT scenarios as per their requirements with the customisation and addition of packages on Node-RED.

Node-RED can interface with Raspberry Pi or Arduino to provide real-time communication with sensors, which can be used for smart agriculture based applications so that farmers can keep track of their fields and crops. Weather forecasting and climate analysis can also be done using such programming platforms. **EFY**

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